

# Supplementary material

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## Abstract

Supporting analyses and tables for the main text; numbers with the prefix S refer to this document.

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## S1. Floor conditioning, chance-constrained training and forecast errors

**One network for all floors.** Regulations and farmers set different floors. The network of the main experiments was trained at  $\rho = 0.9$  only. We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical design step for each floor (Table S1). Its LER exceeds that of the phenology rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037, 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without retraining. A network that received the floor as an additional input and was trained over all four floors did not improve on it (differences in LER  $\leq 0.001$ ). Because the design step already adapts the density to the floor, conditioning the controller on the floor is not needed.

**Layout, rotation limit and harvest-index assumptions.** We checked three assumptions that could drive the ranking (Table S2; 21 pairs of the main split, one training seed). The rotation limit of the trackers sets how much light sharing can deliver: with a limit of 45° the LER of the phenology rule is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking is unchanged. Fixed non-tracking arrays held flat or rotated by 30° about the north–south axis (facing east, not the equator; equator-facing tilts are not modelled) do no better than static trackers (LER 1.048 and 1.043 against 1.053; differences of 0.005 are within the noise of 21 pairs and one training seed), so static trackers are no worse than the fixed arrays we

Table S1: LER of the controllers at different yield-retention floors  $\rho$  (main split; the historical design step is repeated for each floor).

| Method                          | Baseline climate |       |       |       | +3 °C |       |       |       |
|---------------------------------|------------------|-------|-------|-------|-------|-------|-------|-------|
|                                 | 0.80             | 0.85  | 0.90  | 0.95  | 0.80  | 0.85  | 0.90  | 0.95  |
| AV static                       | 1.159            | 1.115 | 1.072 | 1.011 | 1.174 | 1.128 | 1.083 | 1.013 |
| Stress rule                     | 1.380            | 1.282 | 1.182 | 1.077 | 1.364 | 1.269 | 1.172 | 1.071 |
| Phenology rule                  | 1.467            | 1.356 | 1.236 | 1.108 | 1.513 | 1.395 | 1.265 | 1.122 |
| SALS (trained at $\rho = 0.9$ ) | 1.505            | 1.392 | 1.270 | 1.141 | 1.553 | 1.432 | 1.297 | 1.154 |
| SALS conditioned on $\rho$      | 1.504            | 1.391 | 1.269 | 1.141 | 1.553 | 1.431 | 1.296 | 1.154 |

tested; vertical bifacial layouts are not modelled. The ranking of the rules also does not depend on the assumed harvest-index penalty. The window test is weakly informative about the timing of the phenology rule, which shares light through almost the whole growing period, so a shifted window cannot move it much; the informative case is the removal of the penalty. Moving the critical window to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement changes the LER of the phenology rule by at most 0.001, and removing the penalty altogether raises all light-sharing LERs (phenology rule 1.328, SALS 1.353) without changing the order: the phenology rule exceeds seasonal sharing by 0.03 to 0.05 in every case (0.035 with the early window, 0.034 with the late window, 0.053 without penalty), because it produces electricity before canopy development and after maturity, when the crop does not use the light, and not because of the assumed window. The margin of SALS over the written rule changes with these assumptions (0.005 to 0.025 LER), and its margin over the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS equals the tuned rule).

**A floor of 66%.** The German pre-standard DIN SPEC 91434 requires 66% of the reference yield (DIN, 2021). Repeating the design step with  $\rho = 0.66$  (baseline climate, main split, same network trained at 0.9) the ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and SALS 1.71. At this floor the density is limited by the cap and by the mean retention of 0.73 reached at the cap, not by the floor, so the comparison of controllers says little about control; we do not interpret it further.

**Chance-constrained training.** We also trained a network whose ob-

Table S2: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

| Setting                            | Static tracker | Phenology rule | Grid-tuned window | SALS  |
|------------------------------------|----------------|----------------|-------------------|-------|
| Main (limit 60°, window 0.45–0.65) | 1.053          | 1.170          | 1.180             | 1.189 |
| Rotation limit 45°                 | 1.057          | 1.138          | 1.144             | 1.144 |
| Rotation limit 80°                 | 1.053          | 1.206          | 1.219             | 1.224 |
| Window 0.30–0.50                   | 1.053          | 1.169          | 1.176             | 1.185 |
| Window 0.60–0.80                   | 1.053          | 1.171          | 1.181             | 1.182 |
| No harvest-index penalty           | 1.109          | 1.328          | 1.347             | 1.353 |

jective penalises a batch share of seasons below the floor that exceeds 20%. It did not improve the tail. With the mean-based design step its LER was 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-constrained design it reached season-level compliance of 92.6% against 92.0% and an LER of 1.198 against 1.205. The season-level guarantee is obtained by the design step (Section 3.1), not by the training objective.

**Forecast errors.** We raised the forecast errors seen by the controllers to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and of the phenology rule was identical at all four error levels to three decimals (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and 1.416). Neither controller uses the weather forecast: both rely on the crop’s thermal time, which is observable from on-farm temperature records. The stress rule, which depends on the forecast of maximum temperature, degrades with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and its LER from 1.432 to 1.418 as the errors triple.

## S2. Statistical comparison

## S3. Shade-response model, ensembles and economics

## References

- DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – requirements for primary agricultural use. Beuth Verlag, Berlin.
- Laub, M., Pataczek, L., Feuerbacher, A., Zikeli, S., Högy, P., 2022. Contrast-ing yield responses at varying levels of shade suggest different suitability

Table S3: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons).  $\Delta\text{LER}$ : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI;  $p$ : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER;  $\Delta r$ : difference in mean yield retention with its Wilcoxon  $p$ -value.

| Climate  | Baseline         | $\Delta\text{LER}$ [95% CI] | $p$         | % better | $\Delta r$ | $p$         | Sites |
|----------|------------------|-----------------------------|-------------|----------|------------|-------------|-------|
| Baseline | AV static        | +0.205 [+0.176, +0.236]     | $< 10^{-4}$ | 100      | -0.011     | $< 10^{-4}$ | 92    |
| Baseline | Seasonal sharing | +0.088 [+0.068, +0.109]     | $< 10^{-4}$ | 97       | +0.001     | 0.922       | 92    |
| Baseline | Phenology rule   | +0.038 [+0.025, +0.053]     | $< 10^{-4}$ | 84       | +0.001     | 0.723       | 92    |
| Baseline | Stress rule      | +0.096 [+0.071, +0.124]     | $< 10^{-4}$ | 93       | -0.002     | 0.055       | 92    |
| +1.5 °C  | AV static        | +0.208 [+0.180, +0.237]     | $< 10^{-4}$ | 100      | -0.010     | $< 10^{-4}$ | 92    |
| +1.5 °C  | Seasonal sharing | +0.095 [+0.077, +0.114]     | $< 10^{-4}$ | 98       | +0.001     | 0.812       | 92    |
| +1.5 °C  | Phenology rule   | +0.033 [+0.022, +0.045]     | $< 10^{-4}$ | 86       | +0.002     | 0.855       | 92    |
| +1.5 °C  | Stress rule      | +0.108 [+0.083, +0.134]     | $< 10^{-4}$ | 98       | -0.001     | 0.041       | 92    |
| +2 °C    | AV static        | +0.209 [+0.181, +0.238]     | $< 10^{-4}$ | 100      | -0.010     | $< 10^{-4}$ | 92    |
| +2 °C    | Seasonal sharing | +0.097 [+0.079, +0.115]     | $< 10^{-4}$ | 97       | +0.001     | 0.876       | 92    |
| +2 °C    | Phenology rule   | +0.032 [+0.022, +0.044]     | $< 10^{-4}$ | 85       | +0.002     | 0.496       | 92    |
| +2 °C    | Stress rule      | +0.115 [+0.089, +0.142]     | $< 10^{-4}$ | 98       | -0.003     | 0.016       | 92    |
| +3 °C    | AV static        | +0.219 [+0.190, +0.250]     | $< 10^{-4}$ | 100      | -0.009     | $< 10^{-4}$ | 92    |
| +3 °C    | Seasonal sharing | +0.108 [+0.089, +0.129]     | $< 10^{-4}$ | 96       | +0.001     | 0.935       | 92    |
| +3 °C    | Phenology rule   | +0.034 [+0.022, +0.046]     | $< 10^{-4}$ | 86       | +0.002     | 0.330       | 92    |
| +3 °C    | Stress rule      | +0.131 [+0.102, +0.162]     | $< 10^{-4}$ | 98       | -0.002     | 0.142       | 92    |

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Table S4: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with  $s > 0$ ). Covariates: irradiation, climate zone, shade type, intercropping.

| Model                                | RMSE  | MAE   | Median AE | Mean MAE per study |
|--------------------------------------|-------|-------|-----------|--------------------|
| Proportional loss, $y = 1 - s$       | 0.430 | 0.305 | 0.210     | 0.224              |
| One curve for all crops              | 0.372 | 0.265 | 0.193     | 0.208              |
| Curve per crop type, no pooling      | 0.313 | 0.208 | 0.141     | 0.165              |
| Curve per crop type, partial pooling | 0.312 | 0.210 | 0.150     | 0.162              |
| Partial pooling + covariates         | 0.311 | 0.206 | 0.148     | 0.161              |
| Gradient-boosted trees               | 0.321 | 0.219 | 0.162     | 0.184              |

Table S5: Leave-one-study-out prediction error (RMSE of relative yield, observations with  $s > 0$ ) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

| Group                                              | Obs./studies | Proportional | Hierarchical | Boosted trees | Difference | 95% interval     |
|----------------------------------------------------|--------------|--------------|--------------|---------------|------------|------------------|
| C3 cereals                                         | 30/10        | 0.214        | 0.187        | 0.205         | -0.027     | [-0.053, +0.019] |
| Maize                                              | 40/10        | 0.225        | 0.235        | 0.248         | +0.010     | [-0.026, +0.068] |
| Grain legumes                                      | 58/14        | 0.215        | 0.225        | 0.234         | +0.010     | [-0.035, +0.058] |
| Tubers/root crops                                  | 10/2         | 0.250        | 0.254        | 0.257         | +0.004     | [-0.008, +0.053] |
| Four crop types used by the simulator              | 138/30       | 0.220        | 0.222        | 0.234         | +0.002     | [-0.023, +0.030] |
| Other types (berries, forages, fruits, vegetables) | 201/29       | 0.528        | 0.361        | 0.369         | -0.167     | [-0.210, -0.115] |
| All types                                          | 339/58       | 0.430        | 0.312        | 0.321         | -0.118     | [-0.158, -0.071] |

Table S6: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor  $\rho = 0.9$ , split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

| Shade response | Method           | GCR  | $E/E_{PV}$ | Retention | LER  |
|----------------|------------------|------|------------|-----------|------|
| conservative   | AV static        | 0.06 | 0.15       | 0.92      | 1.07 |
|                | Phenology rule   | 0.17 | 0.33       | 0.90      | 1.23 |
|                | Seasonal sharing | 0.16 | 0.28       | 0.90      | 1.18 |
|                | Stress rule      | 0.11 | 0.24       | 0.91      | 1.15 |
| calibrated     | AV static        | 0.07 | 0.18       | 0.91      | 1.09 |
|                | Phenology rule   | 0.19 | 0.37       | 0.90      | 1.28 |
|                | Seasonal sharing | 0.19 | 0.32       | 0.90      | 1.22 |
|                | Stress rule      | 0.13 | 0.29       | 0.91      | 1.20 |
| field-fitted   | AV static        | 0.16 | 0.41       | 0.91      | 1.32 |
|                | Phenology rule   | 0.37 | 0.70       | 0.92      | 1.62 |
|                | Seasonal sharing | 0.37 | 0.61       | 0.92      | 1.53 |
|                | Stress rule      | 0.29 | 0.63       | 0.90      | 1.53 |

Table S7: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean  $\pm$  s.d.): radiation-use compensation  $c$  and multiplier  $h$  on the flowering-window harvest-index penalty (1 = assumed value).

| Crop    | $c$             | $h$ multiplier  |
|---------|-----------------|-----------------|
| Wheat   | $0.91 \pm 0.31$ | $0.74 \pm 0.53$ |
| Rice    | $0.61 \pm 0.25$ | $0.83 \pm 0.33$ |
| Maize   | $0.14 \pm 0.11$ | $0.16 \pm 0.13$ |
| Soybean | $0.05 \pm 0.09$ | $1.66 \pm 0.38$ |
| Potato  | $0.62 \pm 0.20$ | $0.90 \pm 0.50$ |

Table S8: One-at-a-time sensitivity of the net value ( $\text{kUSD ha}^{-1} \text{yr}^{-1}$ , relative to the open field) at  $90 \text{ USD MWh}^{-1}$ , densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

| Parameter                    | Value       | Static arrays | Optimised window | SALS | Conventional plant |
|------------------------------|-------------|---------------|------------------|------|--------------------|
| Base                         |             | 6.3           | 7.3              | 7.6  | 55.2               |
| AV capex ( $\text{USD/Wp}$ ) | 0.7         | 9.7           | 16.6             | 17.1 | 72.8               |
| AV capex ( $\text{USD/Wp}$ ) | 1.5         | 0.6           | -8.2             | -8.2 | 25.8               |
| Discount rate                | 4%          | 8.0           | 12.0             | 12.4 | 64.1               |
| Discount rate                | 8%          | 4.4           | 2.1              | 2.4  | 45.4               |
| Crop prices                  | $0.5\times$ | 6.4           | 7.4              | 7.7  | 57.0               |
| Crop prices                  | $2.0\times$ | 6.0           | 7.0              | 7.3  | 51.6               |
| Electricity price            | 60          | 0.3           | -5.6             | -5.6 | 16.0               |
| Electricity price            | 120         | 12.3          | 20.1             | 20.7 | 94.4               |

Table S9: Design rules outside the ensemble E2 by crop: share of crop–site pairs (%) with mean retention  $\geq 0.9$  under the proportional (P), calibrated (C), field-fitted (F) and harsh (H) responses (phenology rule, three splits pooled). The number of pairs is given in parentheses; with 10 to 20 pairs per crop a difference of one pair is 5 to 10 points. Pooled values are in Table 8.

| Crop (pairs) | Design         | P   | C   | F   | H   |
|--------------|----------------|-----|-----|-----|-----|
| Wheat (17)   | oracle         | 100 | 94  | 100 | 94  |
|              | mean response  | 0   | 0   | 100 | 0   |
|              | 75% of members | 0   | 0   | 100 | 0   |
|              | floor 0.92     | 0   | 0   | 100 | 0   |
|              | floor 0.96     | 12  | 76  | 100 | 0   |
| Rice (20)    | oracle         | 90  | 90  | 80  | 85  |
|              | mean response  | 0   | 0   | 10  | 0   |
|              | 75% of members | 0   | 0   | 40  | 0   |
|              | floor 0.92     | 0   | 10  | 60  | 0   |
|              | floor 0.96     | 95  | 95  | 100 | 85  |
| Maize (17)   | oracle         | 76  | 76  | 71  | 65  |
|              | mean response  | 6   | 6   | 71  | 0   |
|              | 75% of members | 12  | 12  | 82  | 0   |
|              | floor 0.92     | 18  | 18  | 82  | 0   |
|              | floor 0.96     | 71  | 71  | 94  | 53  |
| Soybean (10) | oracle         | 90  | 90  | 90  | 90  |
|              | mean response  | 90  | 90  | 90  | 90  |
|              | 75% of members | 100 | 100 | 100 | 90  |
|              | floor 0.92     | 100 | 100 | 100 | 90  |
|              | floor 0.96     | 100 | 100 | 100 | 100 |
| Potato (12)  | oracle         | 100 | 100 | 100 | 92  |
|              | mean response  | 0   | 0   | 100 | 0   |
|              | 75% of members | 0   | 0   | 100 | 0   |
|              | floor 0.92     | 0   | 0   | 100 | 0   |
|              | floor 0.96     | 42  | 42  | 100 | 42  |